COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 4 — Incremental Model

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Section 11: Classroom Activity — Plan Three Increments

Case Study: Planning staged delivery increments for a departmental Student Attendance Portal (student records, attendance marking, analytical reporting, and low-attendance parent alerts).

1. Increment Breakdown & Staging Plan

Increment	Included Feature Set	Delivered Value & Justification
Increment 1 (Core MVP)	Student profile master data import, subject enrollment, and daily manual attendance marking grid for faculty.	Provides immediate utility; allows faculty to replace paper registers from Week 1 without waiting for advanced analytics.
Increment 2 (Reporting & Visibility)	Attendance percentage computation, student personal dashboard, and monthly shortage reports.	Enables students to monitor their own eligibility and gives HODs visibility into course-wise attendance patterns.
Increment 3 (Automated Alerts & Escalation)	Configurable automated SMS/email alerts for students falling below 75%, and parent notification dispatch.	Completes the operational loop by enforcing compliance rules once data integrity is established in Increments 1 and 2.

2. Architectural & Feature Dependencies

- **Dependency Analysis:** Increment 2 depends directly on Increment 1 because attendance percentage calculation queries daily attendance records stored by Increment 1. Increment 3 depends on Increment 2 because automated alert engines trigger based on threshold calculation routines generated in Increment 2.

3. Pre-Delivery Verification Checks

- **Increment 1 Verification:** Verify database transaction integrity when 30 faculty members submit attendance simultaneously; ensure no duplicate attendance records are written for the same lecture period.
- **Increment 2 Verification:** Validate mathematical accuracy of attendance percentages for edge cases (e.g., student joins late in semester; student has sanctioned medical leaves).
- **Increment 3 Verification:** Verify SMS gateway webhook delivery reports, rate limiting, and handling of invalid parent phone numbers without crashing background queue workers.

4. User Feedback Strategy

- **Faculty Feedback (Post Increment 1):** "How many seconds does it take you to mark attendance for a 60-student class, and is the mobile UI responsive during lecture hall use?"
- **Student Feedback (Post Increment 2):** "Are your subject attendance totals clear, and do you receive immediate updates after a class is logged?"

5. Sequencing Risk

- **Identified Delivery Risk:** If the database schema designed in Increment 1 fails to accommodate elective subject splits or lab batch divisions, refactoring the schema during Increment 2 will break already deployed Increment 1 production data.

Section 14: Homework — Incremental Plan for Online Food Ordering

Proposing a four-stage incremental delivery roadmap for a commercial food ordering application.

Increment	Delivered Features	User Value	Critical Dependency	Mandatory Pre-Delivery Test Checks
Increment 1: Catalog & Cart	Browse restaurant menu, filter items by diet, add items to cart, compute taxes.	Core discovery and bill generation.	None (Baseline)	1. Price calculation with taxes. 2. Cart persistence across sessions.
Increment 2: Checkout & Pay	User login, delivery address selection, payment gateway integration (UPI/Cards).	Enables monetary checkout transaction.	Depends on Inc 1 Cart state	1. Gateway timeout handling. 2. SSL/TLS tokenization verification.
Increment 3: Live Dispatch Tracking	Restaurant kitchen acceptance UI, driver GPS polling, live map route view.	Provides real-time delivery certainty.	Depends on Inc 2 Confirmed Order	1. Driver coordinate update latency. 2. Push notification delivery.
Increment 4: Reviews & Support	Star rating for food/driver, text feedback, automated in-app refund chat bot.	Builds retention and service quality.	Depends on Inc 3 Completed Order	1. Chatbot triage flow. 2. Sentiment score calculation accuracy.

Post-Release Verification Question for Users: "Did you find the real-time driver location accurate to within 100 meters, and did your food arrive within the estimated delivery window?"